

TinySea Bulk Simulation — CSV Column Reference

How to Use

1. Open the template CSV (`bulk_test_template.csv`) in Excel or Google Sheets
2. Each row = one **batch** configuration. Each batch runs `num_scenarios` independent simulations with the same settings but different random seeds
3. Copy a row and change only the values you want to test
4. `batch_name` must be unique per row (it becomes the folder name in the output ZIP)
5. Drag and drop the CSV file into the simulation screen to run all batches automatically
6. Results are downloaded as a ZIP containing per-scenario CSVs and an aggregate summary for each batch

Simulation Settings

Column	Type	Default	Description
<code>batch_name</code>	text	<i>(required)</i>	Label for this batch. Becomes folder name in output ZIP
<code>days</code>	integer	<i>(required)</i>	Number of simulated days per scenario (e.g. 365 = 1 year, 3650 = 10 years)
<code>num_scenarios</code>	integer	<i>(required)</i>	How many times to repeat this configuration with different random seeds

Temperature Settings

All temperatures are in **Celsius**. The simulation converts to Kelvin internally where needed.

Column	Type	Default	Description
<code>base_temp</code>	number	20	Mean annual temperature (C). The center of the seasonal cycle
<code>seasonal_amp</code>	number	10	Seasonal temperature swing (C). Temperature oscillates <code>base_temp</code> +/- this value over a year
<code>climate_trend</code>	number	0	Long-term warming rate (C/year). Set to 0 for stable climate, 1 for +1C per year
<code>variability_mag</code>	number	2	Magnitude of random year-to-year temperature variation
<code>warming_bias</code>	number	1.5	Bias factor that makes warm anomalies more likely than cold ones
<code>daily_var_range</code>	number	5	Daily random temperature fluctuation range (C)
<code>randomness_growth</code>	number	0.5	Rate at which temperature randomness increases over time
<code>autocorrelated</code>	true/false	true	If true, each day's temperature is influenced by the previous day (more realistic). If false, each day is independent

interannual_variation	true/false	true	If true, adds random year-to-year shifts to the base temperature
temp_min	number	-5	Absolute minimum temperature floor (C). Temperature never goes below this
temp_max	number	50	Absolute maximum temperature ceiling (C). Temperature never goes above this

Ecosystem Settings

Column	Type	Default	Description
use_carrying_cap	true/false	true	Enable carrying capacity limit for Tier 1 (prey). When enabled, Tier 1 reproduction slows as population approaches the limit
carrying_cap_t1	integer	5000	Maximum Tier 1 population. Reproduction gradually decreases as population approaches this value (soft cap, not a hard wall)
condition_drain_rate	number	0.15	<i>(Optional)</i> How fast health condition drains when thermal performance is poor. Higher = faster drain
condition_recovery_rate	number	0.10	<i>(Optional)</i> How fast health condition recovers when thermal performance is good. Higher = faster recovery

Species Columns

Each species uses a numbered prefix: `sp1_` for species 1, `sp2_` for species 2, etc. The simulation detects how many species you have by scanning for sequential prefixes. A typical setup has 2 species (sp1 = prey, sp2 = predator).

Identity

Column Suffix	Type	Description
name	text	Species name (e.g. Hexapod, Sheplik). Used in output labels
variant	text	Thermal variant: <code>Common</code> , <code>Tropical</code> , <code>Arctic</code> , or <code>Custom</code> . Determines default thermal curve if Arrhenius values aren't specified
tier	integer	Food chain position: 0 = prey (Tier 1), 1 = predator (Tier 2). Tier 2 eats Tier 1
pop	integer	Starting population count

Reproduction

Reproduction is driven by **Condition** (species health), not instantaneous thermal performance. Condition integrates temperature, feeding, and history — a species with stored health reserves can reproduce even in poor conditions, just at a reduced rate. There is no hard cliff: reproduction scales smoothly from zero (at Condition = 0) to full rate (at Condition = 1.0).

Column Suffix	Type	Default (T1/T2)	Description
repro_mult	number	0.45 / 0.1	Reproduction rate multiplier. Higher = faster population growth. Birth formula: $\text{births} = \text{population} \times \text{reproScale} \times \text{repro_mult}$. The reproScale is condition-based: above threshold it ramps from 0.10 to 1.0, below threshold it ramps from 0 to 0.10. No hard cliff anywhere
repro_thresh	number	0.25 / 0.25	Condition inflection point for reproduction. Above this Condition value, reproduction ramps strongly toward full rate. Below it, reproduction is diminished but non-zero (up to 10% of full rate). Zero reproduction only occurs when Condition = 0 (species is effectively dead). This is NOT a hard cutoff

Death — Condition System

When thermal performance is poor, a species' health **condition** (0-1) drains over time. When condition drops below `death_thresh`, individuals start dying at a rate proportional to how far below the threshold they are.

Column Suffix	Type	Default (T1/T2)	Description
death_thresh	number	0.3 / 0.3	Condition threshold for condition-death. When condition drops below this value, deaths begin. Formula: $\text{severity} = (\text{death_thresh} - \text{condition}) / \text{death_thresh}$
death_rate	number	0.6 / 0.3	Maximum fraction of population killed per day by condition-death (at severity = 1, i.e. condition = 0). Formula: $\text{deaths} = \text{population} \times \text{severity} \times \text{death_rate}$. T1 has higher death rate because prey populations are larger and recover faster

Death — Natural (Background Mortality)

A small constant daily death rate representing old age, disease, and accidents. Applied every day regardless of temperature or condition.

Column Suffix	Type	Default (T1/T2)	Description
natural_death_rate	number	0.02 / 0.01	Daily probability of natural death per individual. T2 (predators) use 0.01 based on allometric scaling (larger animals have lower background mortality)
natural_death_var	number	0.01 / 0.005	Random variance applied to natural death rate each day. Actual rate = $\text{natural_death_rate} \pm \text{random}(\text{natural_death_var})$

Death — Thermal (Instant Kill)

If temperature goes outside a species' absolute survival range, the species dies instantly. This represents lethal temperature extremes.

Column Suffix	Type	Description
ctmin	number	(Optional) Critical thermal minimum (C). Below this temperature, species dies instantly. Default: -5
ctmax	number	(Optional) Critical thermal maximum (C). Above this temperature, species dies instantly. Default: 40

Feeding (Predators Only)

These values are ignored for Tier 1 (prey) species since they don't hunt.

Column Suffix	Type	Default (T2)	Description
eating	number	1.5	Food demand per individual per day (in units of prey). Raw demand = $\text{population} \times \text{eating} \times \text{ThermalPerformance}$
hunt_eff	number	0.75	Base hunting efficiency (0-1). Modified by Holling Type II functional response based on prey-to-predator ratio. At normal ratio (~20:1), efficiency equals this base value. At lower ratios, efficiency drops toward 0. At higher ratios, efficiency approaches 1.0
hunt_var	number	0.15	Random daily variance on hunting success. Actual efficiency = $\text{HollingEfficiency} \pm \text{random}(\text{hunt_var})$

Thermal Performance Curve (Arrhenius Parameters)

These parameters define how well a species performs at different temperatures. The thermal performance curve uses the Arrhenius equation and outputs a value between 0 and p_{max} .

These values are provided by the marine biology team and should not be changed without consultation.

Column Suffix	Type	Description
opt_temp_c	number	Optimal temperature (C) where species performs best
arrhen_breadth	number	Arrhenius breadth parameter. Controls width of the thermal performance curve. Higher = wider tolerance range
arrhen_lower	number	Arrhenius lower deactivation energy. Controls how steeply performance drops below optimal temp
arrhen_upper	number	Arrhenius upper deactivation energy. Controls how steeply performance drops above optimal temp
lower_bound_c	number	Lower thermal performance bound (C). Below this, the Arrhenius curve transitions to decline
upper_bound_c	number	Upper thermal performance bound (C). Above this, the Arrhenius curve transitions to decline

pmax	number	(Optional) Maximum thermal performance (0-1). Caps the Arrhenius curve output. Default varies by variant
temp_offset	number	(Optional) Temperature offset (C) applied to the species' perception of temperature. Default: 0

Default Species Values (Common Variant)

For quick reference, these are the default values used when not specified:

Parameter	Hexapod (Prey, Tier 1)	Sheplik (Predator, Tier 2)
Population	20	4
Eating	0 (doesn't hunt)	1.5
Repro Mult	0.45	0.1
Death Threshold	0.3	0.3
Death Rate	0.6	0.3
Repro Threshold	0.25	0.25
Natural Death Rate	0.02 (2%/day)	0.01 (1%/day)
Natural Death Var	0.01	0.005
Hunting Efficiency	1.0 (ignored)	0.75
Hunting Variance	0 (ignored)	0.15
Optimal Temp	20 C	20 C
Lower/Upper Bound	12 C / 22 C	12 C / 22 C
Pmax	0.9	0.9
CTmin / CTmax	-5 C / 40 C	-5 C / 40 C

Tips

- **A/B testing:** Duplicate a row, change the `batch_name` and one parameter to compare results side by side
- **Keep runs short for iteration:** Use 365 days (1 year) with 3-5 scenarios for quick checks. Use 3650 days (10 years) only for long-term stability tests
- **Large batch warning:** Many batches with high scenario counts and long durations will take significant time. Start small
- **Climate trend = 0:** Set `climate_trend` to 0 to test species in a stable climate (no warming). This isolates seasonal effects from long-term trends
- **Optional columns:** Columns marked optional can be omitted entirely. The simulation will use default values. Old CSVs with removed columns (like `min_deaths`) will still work — unknown columns are ignored with a warning